

Manas Joshi

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Open to relocating - currently Edmonton, AB

SUMMARY

Computing Science 4th-year at the University of Alberta, graduating May 2027, with applied ML across biosignal processing (EEG, EMG), medical-image classification, and production AI deployment. Currently training reinforcement-learning models for prosthetic-limb control at Bionix, building the product-site frontend and business panels at Orbitra, and leading AI and analytics at a Vancouver-based bakery. Starting ECG signal-analysis research on portable cardiac monitoring in September 2026. Seeking AI, ML, or Data Science roles with a bias toward biomedical and biosignal applications.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, C, C++

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, deep learning (CNNs, Transformers), reinforcement learning

Data Science: pandas, NumPy, SciPy, feature engineering, statistical validation, reproducible experiment pipelines

LLM & Agentic AI: Anthropic Claude API, RAG, prompt engineering, response-evaluation loops; agentic AI development via Claude Code / Cursor / Copilot

Production: Docker, FastAPI, Flask, PostgreSQL, Git / GitHub, CI/CD, Linux

EDUCATION

University of Alberta - BSc Computing Science

Expected May 2027

Edmonton, AB, Canada

Relevant Coursework: Algorithms, Machine Learning, Reinforcement Learning, Computer Architecture, Statistics & Probability, Operating Systems

EXPERIENCE

Undergraduate Researcher, ECG Signal Analysis

Incoming Sep 2026

Research group of Prof. Dongsheng Xiao

- **Building an end-to-end ECG analysis pipeline in Python** covering ingestion, filtering, baseline-wander removal, signal-quality assessment, QRS / R-peak detection, and RR-interval analysis (NumPy, SciPy, NeuroKit2, WFDB).
- **Developing abnormal-beat and arrhythmia detection validated against annotated PhysioNet databases**, feeding a long-term monitoring dashboard that summarises event counts, burden, and heart-rate trends across multi-hour recordings.

Frontend & Product Engineering Intern

Jul 2026 - Present

Orbitra Cybersecurity, Remote (orbitrasecurity.com)

- **Build the frontend of the orbitrasecurity.com product site and its business panels** using HTML, CSS, and JavaScript, shipping customer-facing product pages alongside internal business dashboards.
- **Ran security-focused product testing across Orbitra's server infrastructure**, identifying loopholes and consistency gaps across authentication, session handling, and data-flow layers.

Applied ML & Reinforcement Learning Team Member

Oct 2025 - Present

Bionix, Edmonton, AB, Canada

Applied-ML team training and validating reinforcement-learning models that predict prosthetic-limb control from EMG biosignal inputs.

- **Shipped Python RL-control + signal-processing features end-to-end** into the team's shared codebase via Git / GitHub flow with structured peer code review - Agile-style sprint cadence on an interdisciplinary EE / CS / BME team.
- **Validated RL control policies under the RL-Glue framework** on noisy real-world sensor data with pytest-style regression coverage; comfortable adopting AI code assistants daily.

AI & Analytics Lead

Jul 2026 - Present

Take THE Cake Patisserie Ltd., New Westminster, BC (Remote)

- **Strengthened customer connection at scale by deploying an AI customer assistant across 40+ menu, order, and custom-cake inquiry categories**, handing high-touch conversations to staff for a personal moment.
- **Delivered pricing benchmarking across 15+ Metro Vancouver competitors and demand-forecasting from 12+ months of sales**, informing product-line and monthly pricing strategy.

TECHNICAL PROJECTS

NeuroMotion - Real-Time Biosignal ML

Nov 2025

Python, PyTorch, Flask, OpenBCI

Real-time web app that turns streaming brain-signal (EEG) input into predicted user intent under 200 ms (NatHacks 2nd of 40+ teams)

- **Reached 87% validation accuracy at sub-200 ms end-to-end latency on streaming biosignals** (NatHacks 2nd among 40+ teams) by combining sliding-window features with async prediction handling - time-series ML at production tempo.
- **Integrated OpenBCI streaming hardware into a Flask REST backend** with a live JavaScript visualisation frontend and reproducible experiment logging, debugging sub-200 ms latency budgets across the software-to-silicon boundary.

Math-Vision-Model - Image Data ML

Jul 2024 - Jan 2025

Python, TensorFlow, ResNet, EMNIST

Deep-learning classifier that reads handwritten math characters (digits and letters) across 89 symbol classes

- **Reached 88% validation accuracy across 89 handwritten symbol classes** with a ResNet-based image-classification pipeline on the EMNIST dataset.
- **Built a reproducible PyTorch training pipeline with data augmentation, class-balanced sampling, and Scikit-learn evaluation**, enabling rapid iteration on hyperparameter sweeps and architecture variants.